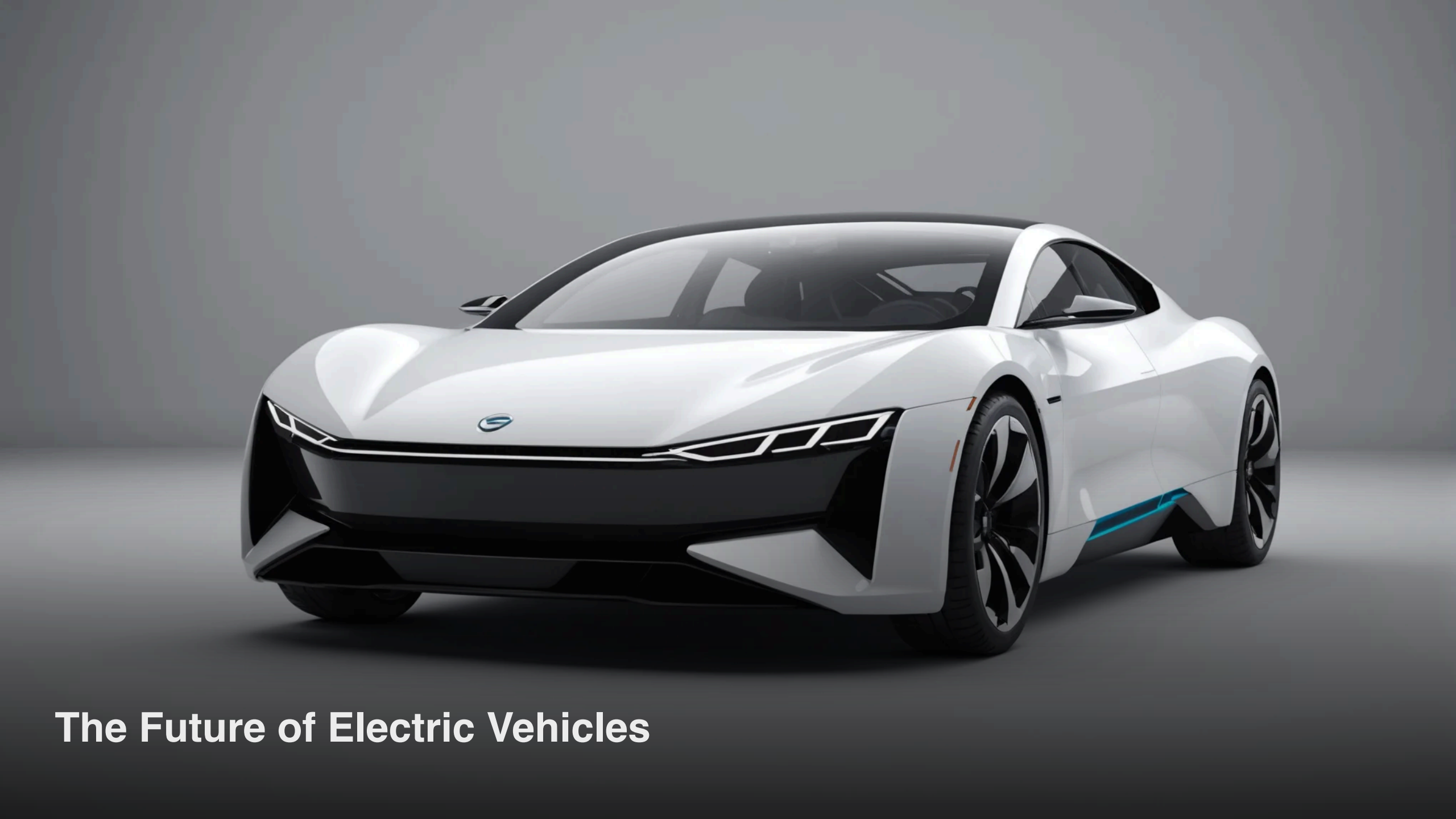


The Future of Electric Vehicles



EV Emissions vs. Conventional Vehicles: Lifecycle and Electricity Mix

Comparing EV and Conventional Vehicle Emissions

Emissions Vary by Country's Electricity Mix

- In countries with clean electricity, such as France and Norway, EVs have significantly lower lifecycle emissions than conventional vehicles[1,^2]
- The emissions advantage of EVs will increase as electricity grids become less carbon-intensive over time[2]
- However, in some regions with a dirtier electricity mix, the climate benefits of EVs may be near-zero or even negative compared to the most efficient conventional vehicles[5]

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Emissions Depend on Real-World Driving Conditions

- Official fuel economy estimates can differ widely from real-world performance, which has a large impact on the comparison between conventional and electric vehicles[4]
- The figures shown in the research adjust emissions for both conventional and electric vehicles to reflect real-world driving conditions rather than test-cycle numbers[3]

Comparing EV and Conventional Vehicle Emissions (Cont.)

Paying Back the "Carbon Debt"

- Even if an EV replaces an existing conventional car, it would still start to cut emissions after less than four years of use compared to continuing to run the older vehicle[6,^7]
- This is because the difference in use-phase emissions is relatively large, with the EV saving two to three tonnes of CO2 equivalent each year in the UK[6]

Additional Benefits of EVs

- Electric vehicles generally have lower maintenance costs, leading to long-term savings for owners[8]
- The shift to electric vehicles contributes to reduced air pollution, improving public health overall[9]
- Adoption of EVs encourages advancements in battery technology and renewable energy utilization[10]

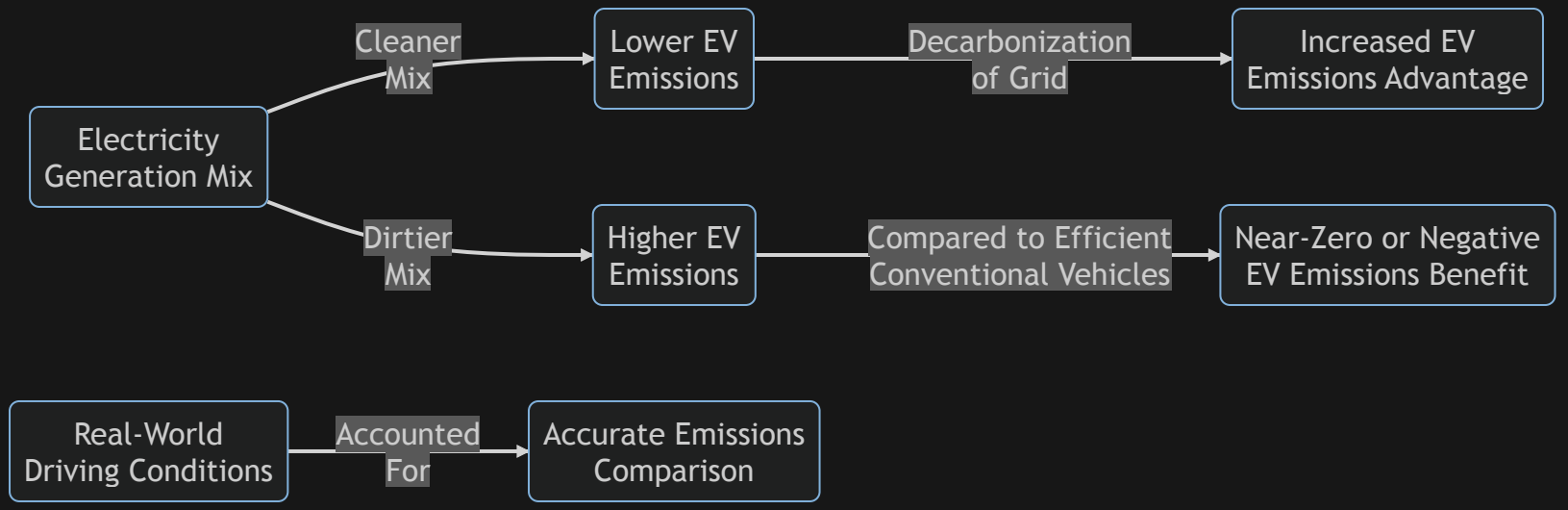
Repaying the Carbon Debt of EV Production

Metric	Nissan Leaf in the UK
Time to "repay" carbon debt from EV production	Less than 4 years[1]
Reason	The difference in use-phase emissions is relatively large, with the EV saving some 2-3 tonnes of CO_2 equivalent each year in the UK.[2] This means that even if a new EV replaces an existing conventional car, it would still start to cut emissions after less than 4 years of use compared to continuing to run the older vehicle.[3]
Assumption	This equation would become even clearer were it not for the generous assumption that the existing conventional car has emissions equal to the average new vehicle.[4]

Importance of Electricity Generation Mix

- The emissions benefits of electric vehicles (EVs) are highly dependent on the local electricity generation mix.[1,^2]
- In regions where the electricity grid is still heavily reliant on fossil fuels, the emissions of EVs may be similar to or even greater than the most efficient conventional vehicles.[3,^4]
- The climate advantages of EVs are much larger in countries like France and Norway, where electricity generation is predominantly from near-zero carbon sources like nuclear and renewables.[1,^2]
- As electricity grids become less carbon-intensive over time, the emissions advantages of EVs will increase further.[2]
- However, the relationship between average emission reductions and marginal emission reductions is not always clear-cut.[5]
- To fully realize the climate benefits of EVs, significant decarbonization of the electricity grid is necessary.[4]
- For the full environmental benefits of electric vehicles to be realized, a strong commitment to transitioning towards renewable energy sources is essential. The electrification of transport must complement broader strategies to reduce greenhouse gas emissions.[4]

Factors Affecting EV Emissions





Tesla Gigafactory Battery Emissions and Estimates

Tesla Gigafactory Battery Emissions

Key Findings

- Estimates of carbon emissions from battery production at Tesla's Gigafactory in Nevada range from 61 to 146 kg CO2-equivalent per kWh of battery capacity[1,^2,^3]
- The lower end of this range (61 kg CO2/kWh) is based on the battery production using renewable energy sources[4]
- The location of the Gigafactory in Nevada, which has a relatively low-carbon electricity grid, helps to reduce the emissions associated with battery manufacturing[5]
- Recent studies have found lower estimates for battery manufacturing emissions compared to earlier research, with an average of around 100 kg CO2/kWh for studies published after 2017[6]

Tesla Gigafactory Battery Emissions (Cont.)

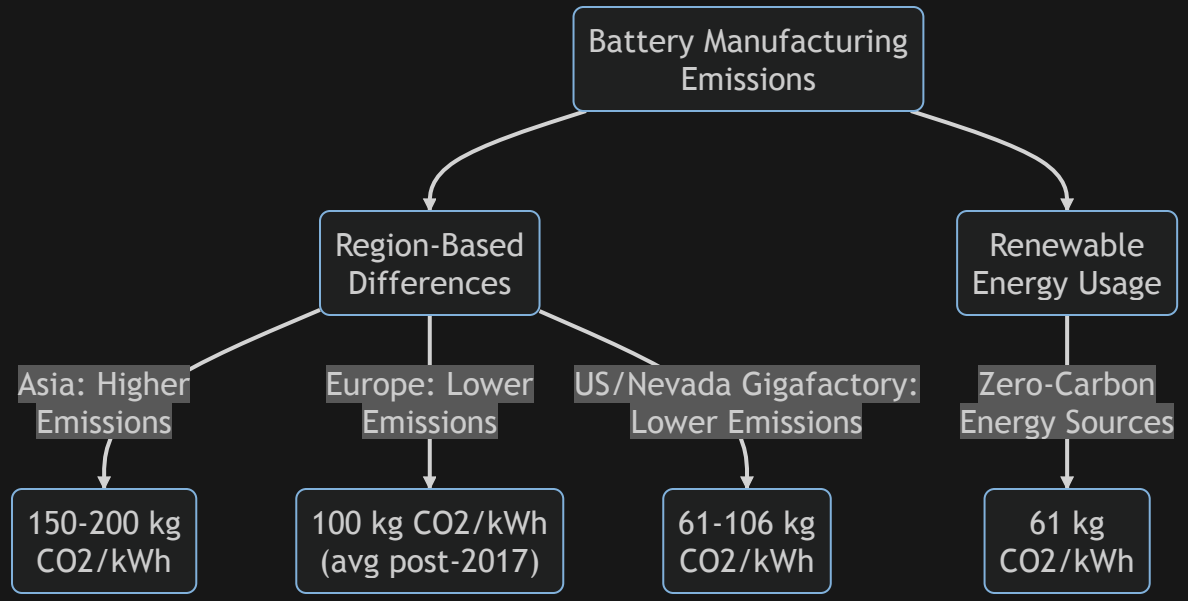
Comparison to Other EVs

- The estimated emissions for the Tesla Model 3 with batteries from the Nevada Gigafactory are similar to the lifecycle emissions of the Nissan Leaf[7]
- This suggests that with the right manufacturing conditions, electric vehicles can have a climate impact comparable to or better than traditional gasoline-powered cars[1]

Comparison of Battery Emission Estimates

Study/Estimate	Battery Emissions (kg CO2e/kWh)
Original IVL Study (2010-2016)	150 - 200 [1]
Revised IVL Study (2019)	61 - 106 (upper bound 146) [2]
Battery Production in Germany (using renewable energy)	62 [3]
Tesla Gigafactory in Nevada (estimated)	61 [4]

Impact of Energy Mix on Battery Emissions



Key Takeaways

Tesla Gigafactory Battery Emissions

- Estimates of emissions from electric vehicle battery manufacturing vary widely, ranging from 61 to 200 kg CO2-equivalent per kWh of battery capacity[1,^2,^3]
- The location of the battery manufacturing facility is a key factor, as regions with lower-carbon electricity grids can significantly reduce battery emissions[4,^5]
- Tesla's Gigafactory in Nevada, which produces batteries for the Model 3, is located in a state with a relatively low-carbon electricity mix, potentially reducing battery emissions to around 61 kg CO2-equivalent per kWh[6,^7]
- The use of renewable energy sources and efficiency measures in battery manufacturing can further lower emissions, with estimates as low as 62 kg CO2-equivalent per kWh in scenarios with 100% renewable energy[8]



Telecommuting and Shared Mobility's Impact on Commuting and Emissions

Telecommuting's Impact on Emissions

Telecommuting reduced commuting emissions during COVID-19

Telework likely to continue, leading to ongoing emissions savings

Shared mobility options like bikes and scooters reduce vehicle use

Shared Mobility's Potential Impact

Telecommuting reduces emissions from commuting

Bike and scooter sharing programs reduce vehicle use

Ridesharing services can lower emissions with zero-emission vehicles

Policies can incentivize ridesharing companies to use cleaner vehicles

Uncertainties Around Shared Mobility

- Shared mobility services like car-sharing and on-demand ride services such as Uber and Lyft have the **potential to reduce emissions**[1] if they:
  - Use high-efficiency or zero-emission vehicles[1]
  - Lean more toward carpooling, with each driver picking up multiple passengers[1]
- However, there is **substantial uncertainty** about the actual impact of these services[1]
- They **might also increase vehicle use and greenhouse gas emissions**[2]
- New policies like the **California Clean Miles Standard** are trying to push companies to use cleaner vehicles and increase passenger loads[3]
  - But it remains to be seen if other regions will adopt similar policies[3]

Comparing Emissions Impacts

Transportation Innovation	Potential Emissions Reduction	Uncertainties	Policy Approaches
Telecommuting	Reduced emissions associated with commuting during the COVID-19 pandemic[1]	Some of the reduction in travel may rebound as the pandemic subsides, but telework is likely to continue for many sectors[2]	N/A
Shared Mobility	Bike and scooter sharing programs can get more people out of vehicles entirely[3]	Car-sharing and on-demand services like Uber and Lyft have the potential to reduce emissions if they use high-efficiency or zero-emission vehicles, or if their services lean more toward carpooling, but there is substantial uncertainty about the impact of these services[4]	The California Clean Miles Standard is helping to push companies like Uber and Lyft to use cleaner vehicles and increase their passenger loads, though it remains to be seen whether other regions will adopt similar policies[5]
Electric Vehicles	A major, rapid rethinking of how people get around globally would be required to achieve drastic emissions reductions[6]	N/A	N/A



Electric Vehicles' Emissions and Climate Impact

The Environmental Impact of Electric Vehicles

Key Takeaways:

- Electric vehicles (EVs) have significantly lower emissions than gas-powered cars, even when accounting for the emissions from electricity generation.[1,^2]
- While the manufacturing process of EV batteries is more carbon-intensive, this is offset by their cleaner operation over the vehicle's lifetime.[1,^2]
- EVs play a crucial role in combating climate change by reducing greenhouse gas emissions from the transportation sector, which is a major contributor to global emissions.[1,^2] **Environmental Benefits of EVs:**
- EVs produce zero direct emissions during operation, unlike gas-powered cars that emit greenhouse gases[1,^2]
- As the electricity grid becomes cleaner, the emissions associated with charging EVs will continue to decrease[1]
- EVs are more energy-efficient than gas-powered cars, requiring less energy to travel the same distance[1,^2]

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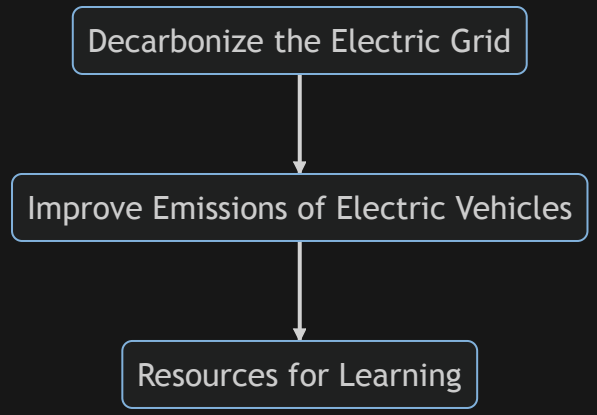
The Environmental Impact of Electric Vehicles (Cont.)

- The manufacturing process of EV batteries is more carbon-intensive than gas-powered car production.[1,^2]
- However, the emissions from battery manufacturing are offset by the significantly lower emissions during the vehicle's lifetime.[1,^2]
- Advancements in battery technology and renewable energy sources for manufacturing can further reduce the carbon footprint of EV batteries.[1,^2] **EVs' Role in Climate Change Mitigation:**
- Transportation is a major contributor to global greenhouse gas emissions, accounting for about 24% of total emissions.[1,^2]
- Widespread adoption of EVs can help reduce emissions from the transportation sector, making them a crucial part of the solution to combat climate change.[1,^2]
- As the electricity grid becomes cleaner, the environmental benefits of EVs will continue to grow.[1,^2]
- Increasing the efficiency of energy sources used in EVs contributes to overall emission reductions in the sector and aligns with global targets for a sustainable future.[1,^2]

Emissions Comparison: Electric vs. Gas-Powered Vehicles

Emissions Metric	Electric Vehicles	Gas-Powered Vehicles
Manufacturing Emissions	**Higher** due to battery production[1]	**Lower**
Fuel/Electricity Production Emissions	**Lower** as electricity grid becomes cleaner[1]	**Higher** due to gasoline production
Vehicle Operation Emissions	**Zero** direct emissions[1]	**Higher** direct emissions from combustion
Total Lifecycle Emissions	**Better** for the climate, especially as the grid decarbonizes[1]	**Worse** for the climate

The Path to a Cleaner Electric Grid



Key Takeaways

Electric vehicles have lower carbon emissions than gas cars

Decarbonizing the grid further improves EVs' climate benefits

EVs play a crucial role in reducing transportation emissions



Alternative Fuels for Heavy Transportation

Challenges in Electrifying Heavy Transportation

Size and weight of batteries limit electrification of heavy vehicles

Limited range of electric trucks and ships hinders adoption

Lack of affordable, scalable alt. fuel options for heavy transport

Exploring Alternative Fuels for Heavy Vehicles

Challenges with Electrifying Heavy Transportation

- Larger and heavier vehicles like trucks, ships, and airplanes are more difficult to electrify due to the size and weight of the batteries required[1,^2]
- The cost and feasibility of alternative fuel technologies for these heavy vehicles is still a major challenge[2,^3]

Exploring Alternative Fuels for Heavy Vehicles (Cont.)

Alternative Fuel Options for Heavy Vehicles

1. **Hydrogen**:
    - Hydrogen fuel cells are being explored as a potential zero-emission solution for heavy-duty trucks, ships, and airplanes[2,^3]
    - However, substantial advances in the technology are still needed to make hydrogen economically feasible and widely adopted[3]

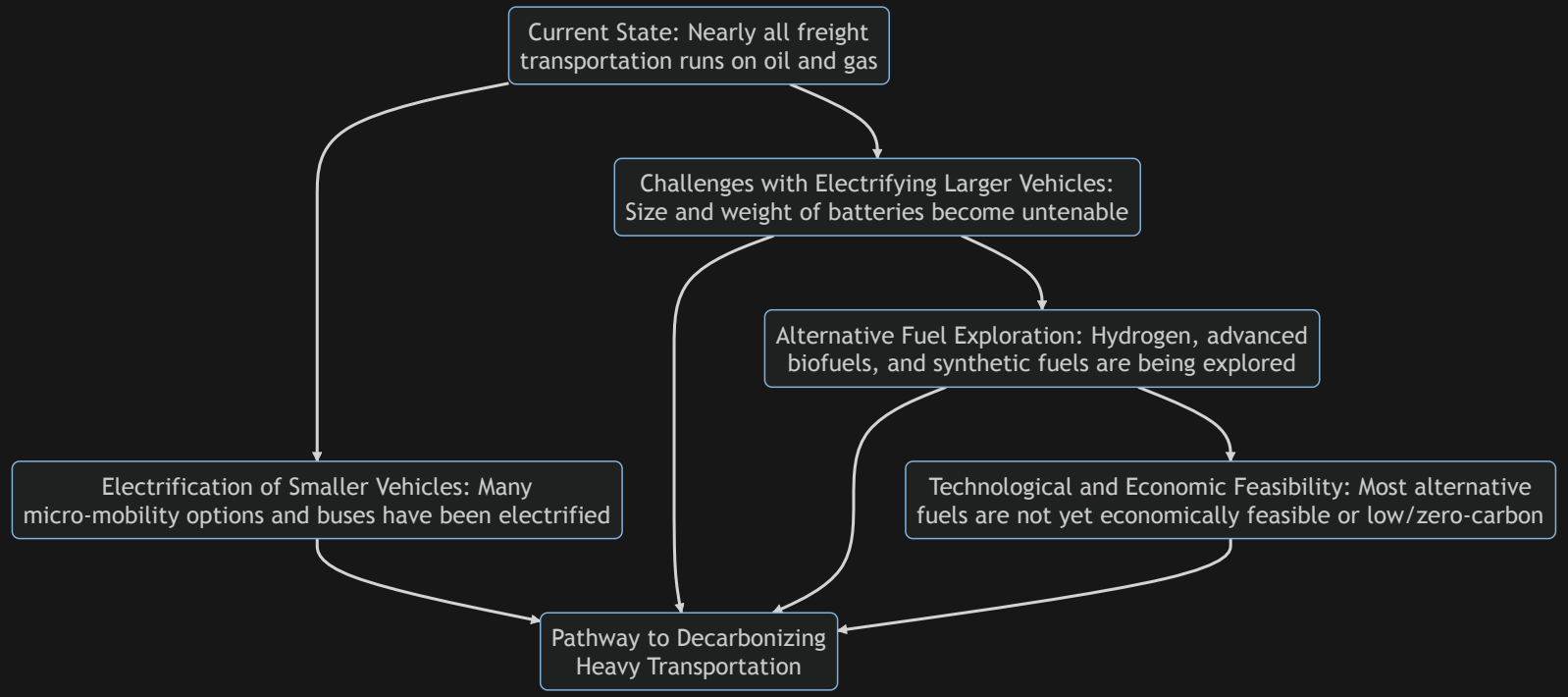
Exploring Alternative Fuels for Heavy Vehicles (Cont.)

2. **Advanced Biofuels**:
    - Biofuels derived from non-food biomass sources are being investigated as a replacement for fossil fuels in heavy transportation[2,^3]
    - These advanced biofuels have the potential to be low- or zero-carbon, but significant technological breakthroughs are still required[3]
3. **Synthetic Fuels**:
    - Synthetic fuels produced from renewable energy sources and captured carbon dioxide are another alternative being explored[2,^3]
    - Like advanced biofuels, synthetic fuels face challenges in becoming economically viable and widely available[3]

Comparison of Alternative Fuel Options

Fuel Type	Energy Density	Emissions Profile	Cost	Technological Readiness
Hydrogen	High energy density, but requires large storage tanks.[1]	*Zero direct emissions*, but production process can have emissions.[1]	Currently more expensive than fossil fuels, but costs expected to decrease.[1]	Significant advances still needed to ensure low/zero-carbon production.[1]
Advanced Biofuels	Moderate energy density, requires less storage space than hydrogen.[2]	*Potential for low/zero emissions*, depending on feedstock and production process.[2]	More expensive than fossil fuels, but costs may decrease with scale.[2]	Substantial advances in technology still required to become economically feasible.[2]
Synthetic Fuels	Moderate energy density, requires less storage space than hydrogen.[3]	*Potential for low/zero emissions*, depending on production process.[3]	More expensive than fossil fuels, but costs may decrease with scale.[3]	Significant technological advances needed to ensure low/zero-carbon production.[3]
Fossil Fuels	High energy density, well-established infrastructure.[4]	*High emissions*, major contributor to climate change.[4]	Relatively low cost, but price volatility.[4]	Mature technology, but unsustainable long-term.[4]

Pathway to Decarbonizing Heavy Transportation





IPCC Report on Transportation Emissions Reduction Targets by 2050 and

IPCC Emissions Reduction Targets for Transportation by 2050

50% decrease in transportation carbon intensity by 2050

91% decrease in transportation carbon intensity by 2100

Combine transportation improvements with cleaner electricity grid

Pathways to Achieve IPCC Transportation Emissions Targets

Accelerating Adoption of Electric Vehicles

- Electric vehicles (EVs) could play a key role in reducing transportation emissions by up to 91% by 2100 when combined with a cleaner electricity grid[1]
- Scenarios indicate the carbon intensity of the transportation sector would need to decrease by about 50% by 2050 to stay within the 1.5°C warming target[2]
- Rapid transition to EVs, along with other innovations, is crucial to achieving these emissions reduction targets[3]

Improving Public Transportation and Active Mobility

- Increased investment in public transportation, walking, and cycling infrastructure can significantly lower transportation emissions[4]
- Integrating public transit with other modes of transportation, such as electric bikes and scooters, can make low-emission travel more accessible[5]

Pathways to Achieve IPCC Transportation Emissions Targets (Cont.)

Advancing Biofuel and Alternative Fuel Technologies

- Sustainable biofuels and other alternative fuels can help decarbonize hard-to-electrify transportation sectors, such as aviation and heavy-duty vehicles[6]
- Continued research and development in these technologies is necessary to scale up production and reduce costs[7]

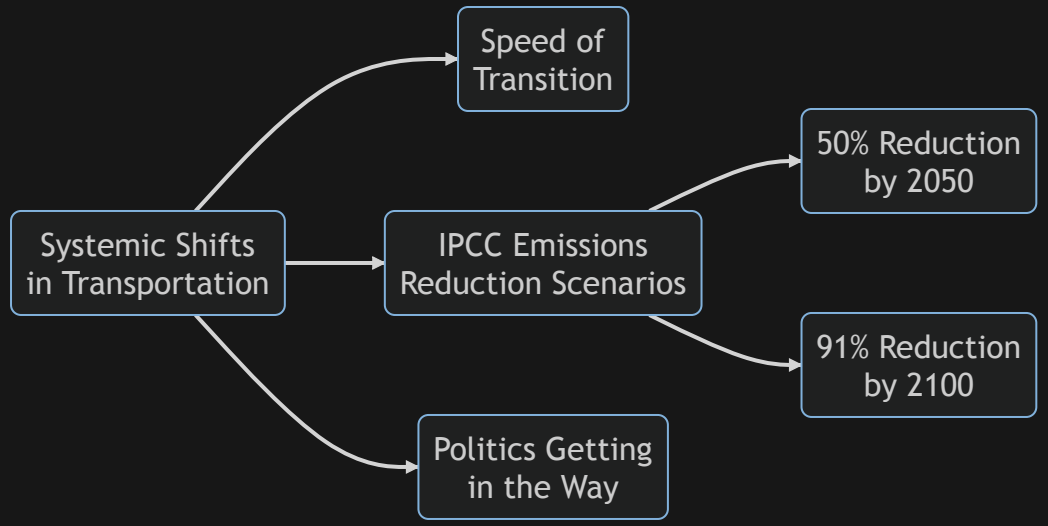
Supportive Policies and Infrastructure

- Coordinated policy changes, such as carbon pricing, vehicle emission standards, and investment in charging/fueling infrastructure, are crucial to driving the transition to low-emission transportation[8]
- Overcoming political barriers and aligning stakeholders will be key to implementing these systemic changes in the transportation sector[9]

IPCC Emissions Reduction Scenarios for Transportation

Emissions Reduction Scenario	Carbon Intensity Reduction by 2050	Carbon Intensity Reduction by 2100
Optimistic Scenario	Decrease by **about 50%**[1]	Decrease by **as much as 91%**[2]
Factors Underlying Scenarios	- **Rapid adoption of transportation innovations**[3] - **Decarbonization of the electricity grid**[2]	- **Rapid adoption of transportation innovations**[3] - **Decarbonization of the electricity grid**[2]

Factors Influencing Transportation Emissions Reductions





IPCC Report on Renewable Energy and EV Costs Reducing Transportation Emissions by

IPCC Report on Renewable Energy and EV Costs

Falling costs of renewable energy and EV batteries

Rising adoption of renewable energy and EVs

Potential 80-90% reduction in transportation emissions by 2050

How Advances in Renewable Energy and EVs Reduce Emissions

Falling Costs and Rising Adoption

- Costs are falling for key forms of renewable energy and EV batteries[1,^2]
- Adoption of these technologies is rising[1,^2]

Emission Reduction Potential

- In the most aggressive scenarios, transportation emissions could be reduced by 80-90% of current levels by 2050[3]
- This is due to "transportation transformations – some just starting and others expanding"[3]

How Advances in Renewable Energy and EVs Reduce Emissions (Cont.)

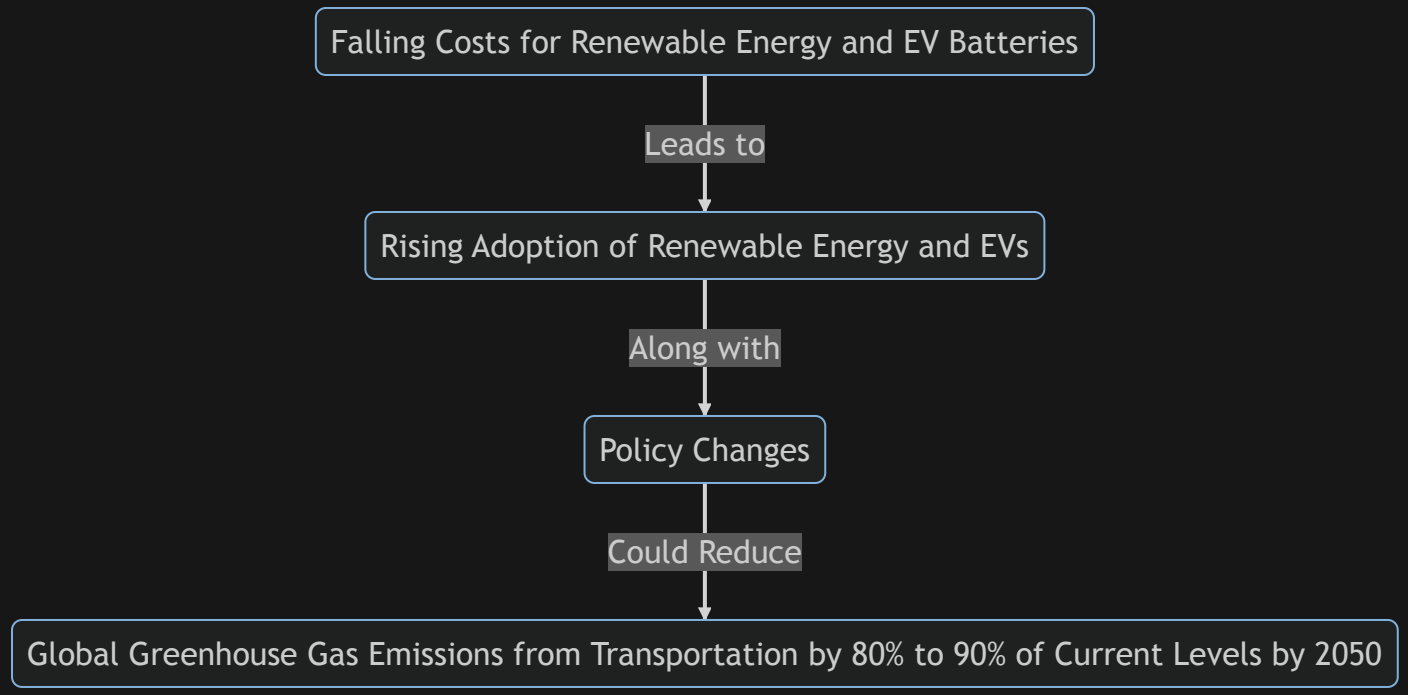
Role in Combating Climate Change

- The IPCC report concludes that these advances in renewable energy and EVs, along with policy changes, have **slowed the growth of climate change in the past decade**[4]
- However, **deep, immediate cuts are necessary to stop emissions growth entirely and keep global warming in check**[4]

IPCC Emission Reduction Scenarios for Transportation

Emission Reduction Scenario	Percentage Reduction by 2050	Key Factors
Most Aggressive Scenario	80% to 90% reduction[1]	- Rapid adoption of electric vehicles - Widespread use of renewable energy in transportation[2]
Moderate Scenario	Significant but unspecified reduction[3]	- Increasing adoption of electric vehicles - Continued growth in renewable energy use[4]
Conservative Scenario	Slower emissions reductions[5]	- Gradual transition to electric vehicles - Moderate growth in renewable energy use[6]

The Pathway to Reduced Transportation Emissions





Lithium Batteries and Energy Storage

What are Lithium-Ion Batteries?

High energy density for their weight

Can be recharged many times

Widely used in electric vehicles and grid storage

Lithium Mining and Environmental Impact

Lithium Extraction Methods

- Lithium is found in rock ores, which are mined and crushed[1]
- Lithium can also be extracted from briny water through an evaporation process[1]

Major Lithium Reserves

- Major lithium reserves are located in countries like Chile, Australia, and China[2]
- These regions have ideal geological conditions for lithium extraction, such as salt flats and brine deposits[2]

Lithium Mining and Environmental Impact (Cont.)

Environmental Impacts of Lithium Mining

- Lithium mining can have significant environmental impacts, including:
  - High water usage, especially in arid regions where many lithium reserves are located[1]
  - Land disturbance and habitat destruction from mining operations[1]
  - Chemical pollution from the extraction and processing of lithium[1] ## Mitigating Environmental Impacts
- Efforts are being made to develop more sustainable lithium extraction methods, such as using renewable energy sources and improving water recycling[3]
- Recycling of lithium-ion batteries can also help reduce the need for new lithium extraction[3]

Comparing Lithium-Ion Batteries to Other Energy Storage Technologies

Technology	Energy Density	Power Density	Cycle Life	Cost	Environmental Impact
Lithium-Ion Batteries	Hold a lot of energy for their weight[1]	Can provide the power to run heavy machinery[1]	Can be recharged many times[1]	-	Emissions from mining and manufacturing, but savings when driving[2]
Lead-Acid Batteries	Lower energy density than lithium-ion[1]	Lower power density than lithium-ion[1]	Shorter cycle life than lithium-ion[1]	Lower cost than lithium-ion	Contain lead, which can be harmful if not disposed of properly
Flow Batteries	Lower energy density than lithium-ion[1]	Lower power density than lithium-ion[1]	Longer cycle life than lithium-ion[1]	Higher cost than lithium-ion	Use non-toxic, non-flammable electrolytes, reducing environmental impact
Hydrogen Fuel Cells	Higher energy density than lithium-ion[1]	Higher power density than lithium-ion[1]	Longer cycle life than lithium-ion[1]	Higher cost than lithium-ion	Produce water as the only byproduct, reducing environmental impact

The Role of Lithium-Ion Batteries in the Transition to Renewable Energy

